



Capacitive Soil Moisture Sensor (v1.2)

Hardware Overview



What is it?

This **Capacitive Soil Moisture Sensor** is a robust, corrosion-resistant sensor used to measure how much water is in soil or dirt. Unlike older, cheaper sensors that pass electricity directly through the soil (which causes them to rust and dissolve in days), this sensor acts like a giant capacitor, making it practically indestructible in normal use.



Electrical Explanation

The sensor uses a specialized high-frequency circuit to detect moisture without ever exposing its metal parts to the water.

- Capacitance: The two black prongs pushed into the dirt create an invisible electromagnetic field. Water has a very high dielectric constant, meaning it heavily disrupts this magnetic field.
- Analog conversion: A microchip on the board (a 555 timer) measures that disruption and turns it into a raw analog voltage. The more water there is, the lower the voltage output!

Key Hardware Facts

- **Connection**: Analog Pin (e.g., A3).
- **Voltage**: 5V
- **Signal Flow**: Ambient Moisture → PCB Traces (Capacitor) → 555 Timer IC → Analog Voltage → Arduino ADC → Processor.

Software & Programming



kcomp_SoilSensor.h

This library simplifies the analog reading process, smooths out electrical noise, and handles the inverted mathematical mapping required to get a clean 0–100% reading. It further supports software-sided calibration.

Usage: `#include <kcomp_SoilSensor.h>`

Function Reference

Function Name	Description & Parameters
<code>void initSoilSensor(uint8_t pin)</code>	Setup: Initializes the specified analog pin (e.g., A3) as an input. Must be called once in <code>setup()</code> .
<code>void setSoilCalibration(uint16_t dry_value, uint16_t wet_value)</code>	Calibrate: Maps the sensor's unique raw voltages to 0% and 100%. Pass in the raw number when dry, and the raw number when submerged. Calling <code>setSoilCalibration(560, 220);</code> should give acceptable results.

<code>uint16_t readSoilRaw()</code>	Read Raw: Returns the raw 10-bit analog voltage reading (0–1023) directly from the Arduino Analog Digital Converter (ADC).
<code>uint8_t readSoilMoisture()</code>	Read Moisture: Calculates and returns the calibrated moisture level as a percentage (0% to 100%).

⚠ Pitfalls & Troubleshooting

- The “Inverted Logic” Trap

One naturally assumes that more water equals a higher number. This sensor does the exact opposite.

The Fix: Because of how the internal timer circuit works, dry air yields a high voltage, and being submerged in water yields a low voltage. The library automatically handles this backwards math using the calibration values you provide!

- The Calibration Drift

Because these are budget hardware modules, the manufacturing tolerances vary wildly. Two identical sensors can have completely different baseline readings.

The Fix: Do not trust “textbook” numbers. You must physically test your specific sensor in dry air and a glass of water to find its unique values, then enter them into `setSoilCalibration()`.

📄 Sample Program

This program initializes the sensor using custom calibration values and prints the raw data alongside the human-readable percentage every second.

```
#include <kcomp_SoilSensor.h>

void setup() {
  Serial.begin(9600);
  // 1. Initialize the sensor on Analog Pin 0
  initSoilSensor(A0);
  // 2. Set your custom calibration values!
  // Example Values: Dry Air = 560, Glass of Water = 220
  setSoilCalibration(560, 220);
  Serial.println("K-Comp Soil Monitor Online!");
}

void loop() {
  // 3. Read the Raw Hardware Value
  uint16_t raw_val = readSoilRaw();
  // 4. Read the Calculated Percentage
  uint8_t moisture = readSoilMoisture();
  // 5. Output results
  Serial.print("Raw Analog: ");
  Serial.print(raw_val);
  Serial.print(" | Moisture: ");
  Serial.print(moisture);
  Serial.println(" %");
  // Wait 1 second before checking again
  delay(1000);
}
```

💡 **Try it out:** You can find the full sample program

SimpleCapacitiveSoilMoistureSensor in the K-Comp examples folder!